

2016.0 RANGE ROVER (LG), 204-06

RIDE AND HANDLING OPTIMIZATION

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

Ride and handling optimization incorporates the terrain response system which links a number of modules around the vehicle to give the best combination of settings in the different systems

For a detailed description of the ride and handling system and operation, refer to the relevant description and operation section of the workshop manual.

REFER to: [Ride and Handling Optimization](#) (204-06 Ride and Handling Optimization, Description and Operation).

INSPECTION AND VERIFICATION

WARNING:

Before carrying out a road test, make sure the vehicle is safe to do so.
Failure to follow this instruction may result in personal injury

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests

1. Verify the customer concern
2. Visually inspect for obvious signs of mechanical or electrical damage

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Tire condition, pressures, etc ▪ Driveline components (correct installation, damage, etc) ▪ Engine components (correct installation, damage, etc) ▪ Transmission components (correct installation, damage, etc) ▪ Suspension components (correct installation, damage, etc) ▪ Dynamic response system fluid level/condition, pipes, reservoir, etc ▪ Dynamic response hydraulic pump ▪ Drive belt condition ▪ Hoses ▪ Dynamic response valve block ▪ Accelerometers (correct fitment, etc)	▪ Fuses ▪ Harnesses/Connectors ▪ Terrain response switchpack ▪ Integrated suspension control module ▪ Engine control module ▪ Transmission control module ▪ Transfer case control module ▪ ABS control module ▪ Rear differential control module ▪ Dynamic response control module ▪ CAN circuits ▪ Upper accelerometer

- Lower accelerometer
- Steering angle sensor module
- Instrument cluster

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the symptom chart, alternatively check for DTCs and refer to the DTC index

SYMPTOM CHART

Because the overall function of the system is dependent on sub-systems, it is possible to misinterpret displays in the message centre as being terrain response faults when they are actually a result of a fault in one of the sub-systems

Refer to the table below for help in deciding when to investigate terrain response faults and when the fault is likely to be in a sub-system

SYMPTOM	DESCRIPTION	POSSIBLE CAUSES	ACTION
Message centre display indicating a sub-system fault	The message centre indicates to the driver that a fault has occurred and in which sub-system	▪ Any sub-system fault supported by the message centre	For details of the available messages, refer to the relevant section of the workshop manual. Carry out a complete vehicle DTC read and follow the diagnostic routine(s) indicated
Message centre display: System fault special programs not available , terrain response switch operation	This message will display when a sub-system fault has occurred if the driver attempts to change the special program, and at each ignition on cycle for 5 seconds until the fault is rectified	▪ Any sub-system fault supported by the message centre	For details of the available messages, refer to the relevant section of the workshop manual. Carry out a complete vehicle DTC read and follow the diagnostic routine(s) indicated

normal			
Message centre display: System fault special programs not available , ALL terrain response switch LEDs illuminated	CAN circuit errors	▪ CAN circuit: short circuit to ground ▪ CAN circuit: short circuit to power ▪ CAN circuit: high resistance	Carry out a complete vehicle DTC read and follow the diagnostic routine(s) indicated
Special program changes not available	User error	▪ Engine not running ▪ Rock crawl selected with transfer box in high range ▪ Special program change attempted with ABS or dynamic stability control active ▪ Special program change attempted with an overheat condition present in the centre or rear differential	Refer to the relevant section of the workshop manual. Make sure that the driver is familiar with the correct operation of the system

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Poor on-center response	▪ System bleed required ▪ Roll bar bushes ▪ Steering angle sensor offset ▪ Control module adaptive data	Carry out the manual bleed procedure. REFER to: Active Stabilization System Bleeding (204-06 Ride and Handling Optimization, General Procedures). Check the roll bar bushes. Check and calibrate the steering angle sensor, clear the adaptive data after calibration. Run hydraulic self-test routine and check response
Asymmetrical response	▪ Steering angle sensor offset ▪ Accelerometer calibration ▪ System bleed required	Check and calibrate the steering angle sensor, clear the adaptive data after calibration. Calibrate the accelerometers using the approved diagnostic system. Carry out the manual bleed procedure. REFER to: Active Stabilization System Bleeding (204-06 Ride and Handling Optimization, General Procedures). Run hydraulic self-test routine and check response
Excessive roll	▪ System bleed required ▪ Roll bar bushes ▪ Stabilization links	Carry out the manual bleed procedure. REFER to: Active Stabilization System Bleeding (204-06 Ride and Handling Optimization, General Procedures). Check the roll bar bushes and stabilization links. Run hydraulic self-test routine and check response
Powered roll-rock	▪ Harness faults ▪ Valve block fault ▪ Control module adaptive data ▪ Accelerometer fault	Check for DTCs indicating any of the possible causes are present. Run hydraulic self-test routine and check response
Oversteer or understeer	▪ Stabilization links ▪ System actuators	Check the stabilization links and the system actuators. Run hydraulic self-test routine and check response

DTC INDEX

For a list of diagnostic trouble codes (DTCs) that could be logged on this vehicle, please refer to section 100-00. REFER to: (100-00 General Information)

[Diagnostic Trouble Code Index - DTC: Dynamic Response Control Module \(DRM\)](#) (Description and Operation),

[Diagnostic Trouble Code Index: Terrain Response Switchpack](#) (Description and Operation).